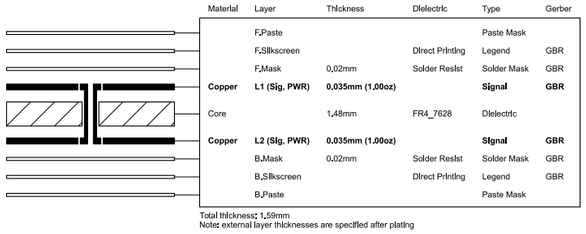
BATTERY_INTERFACE_BOARD Fabrication Document

Layer Stack Legend



Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	**L1 (Sig, PWR)**	**0.035mm (1.00oz)**		**Signal**	**GBR**
Core		1.48mm	FR4_7628	Dielectric	
Copper	**L2 (Sig, PWR)**	**0.035mm (1.00oz)**		**Signal**	**GBR**
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1.59mm
Notes: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0.2032	0.28	L2

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: IMMERSION GOLD

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLUE.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE, AND PROCESS BY A PCB FABRICATOR APPROVED BY BLUE DEVIL MOTOR SPORTS.

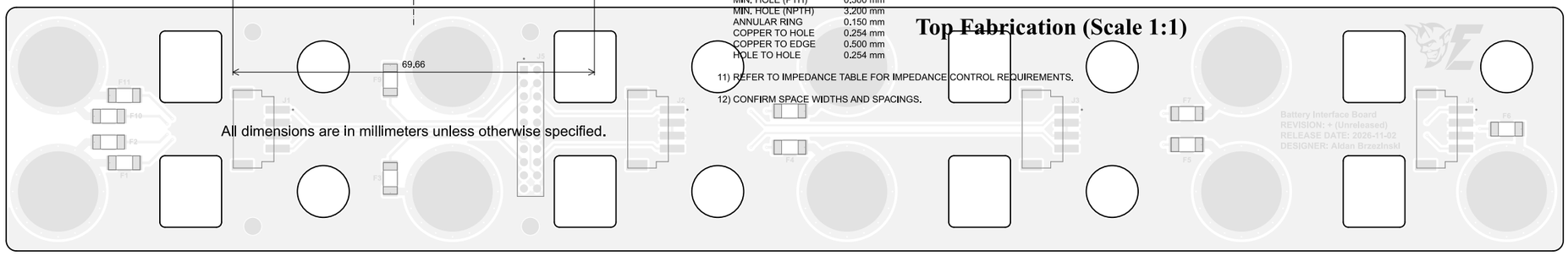
10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	300.000 × 47.000 mm
BOARD THICKNESS	1.590 mm
TRACE WIDTH	0.400 mm
TRACE TO TRACE	0.200 mm
MIN. HOLE (PTH)	0.300 mm
MIN. HOLE (NPTH)	3.200 mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.254 mm
COPPER TO EDGE	0.500 mm
HOLE TO HOLE	0.254 mm

11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.

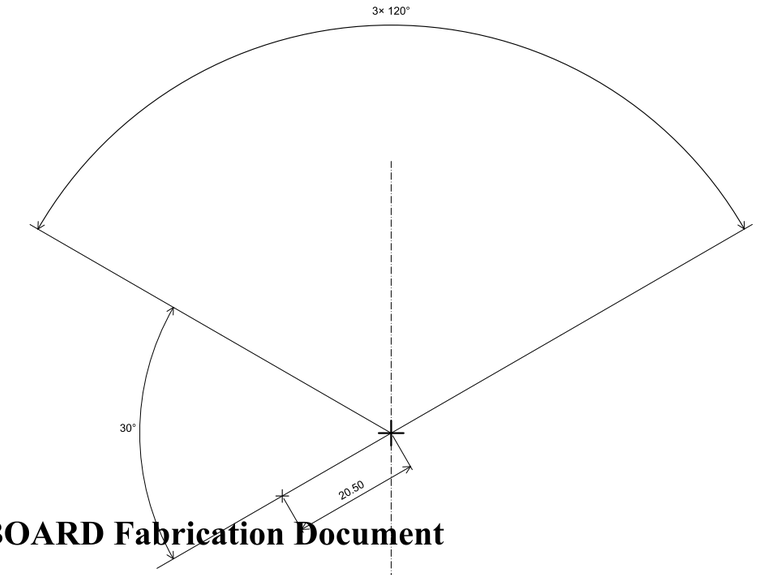
12) CONFIRM SPACE WIDTHS AND SPACINGS.

Top Fabrication (Scale 1:1)

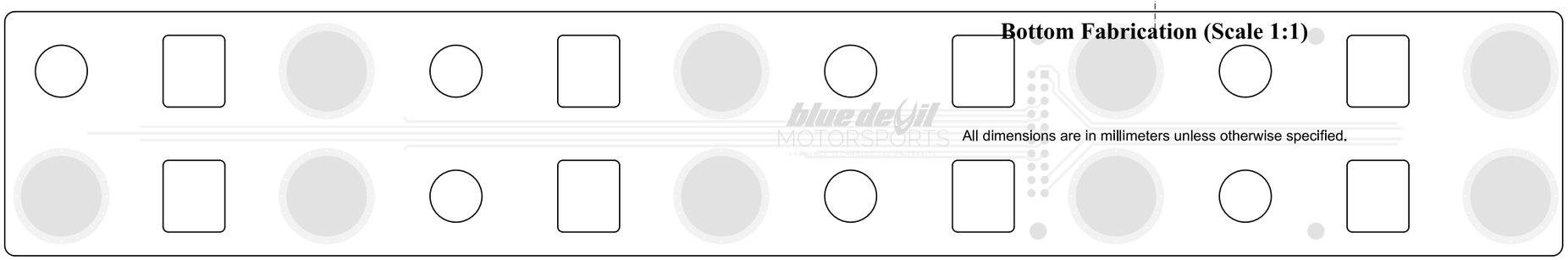


All dimensions are in millimeters unless otherwise specified.

Comments:	Company: **Blue Devil Motor Sports**			Variant: CHECKED	Git Hash: c247924
	Board Name: **BATTERY_INTERFACE_BOARD**			Project Name: **Battery Interface Board**	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: battery_interface_board.kicad_pcb	Designer: Aidan Brzezinski		Date: 2026-10-12	Revision: + (Unreleased)
Sheet Path:		Reviewer:		Size: **A3**	Sheet: **1** of **8**



BATTERY_INTERFACE_BOARD Fabrication Document



Bottom Fabrication (Scale 1:1)

All dimensions are in millimeters unless otherwise specified.

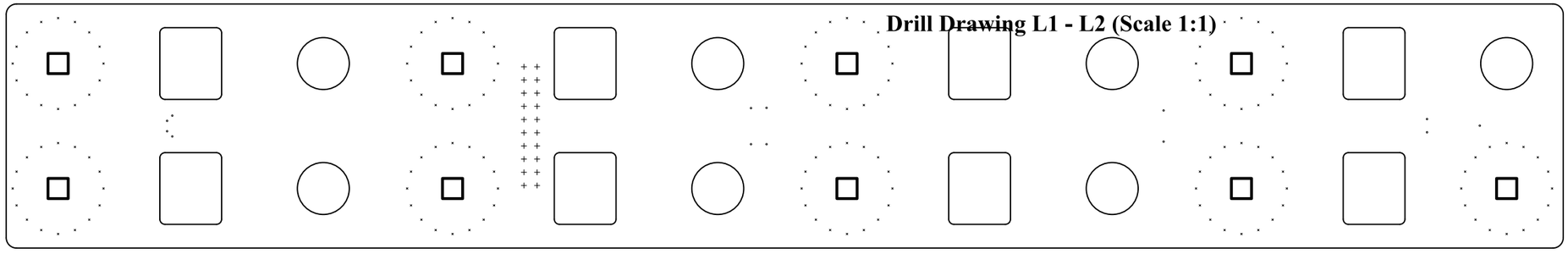
Comments:	Company: **Blue Devil Motor Sports**		Variant: CHECKED	Git Hash: e247924
	Board Name: **BATTERY_INTERFACE_BOARD**		Project Name: **Battery Interface Board**	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: battery_interface_board.kicad_pcb	Designer: Aidan Brzezinski	Date: 2026-10-12	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **2** of **8**

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	144	0.30mm (11.81mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Pad
○	13	0.30mm (11.81mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Via
+	20	1.02mm (40.16mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Pad
□	9	5.50mm (216.54mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Pad
	Total 186					

BATTERY_INTERFACE_BOARD Fabrication Document

Drill Drawing L1 - L2 (Scale 1:1)



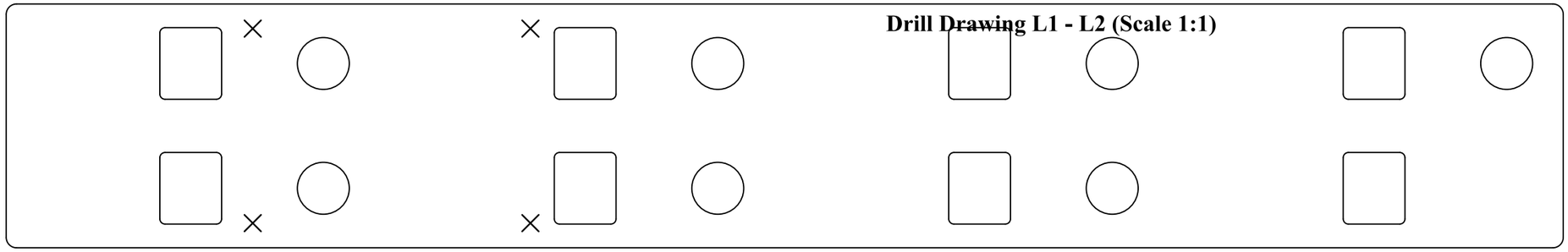
Comments:	Company: Blue Devil Motor Sports		Variant: CHECKED	Git Hash: e247924
	Board Name: BATTERY_INTERFACE_BOARD		Project Name: Battery Interface Board	
Sheet Title: Drill Drawing (L1 - L2)	File Name: battery_interface_board.kicad_pcb	Designer: Aidan Brzezinski	Date: 2026-10-12	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: A3	Sheet: 3 of 8

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	4	3.20mm (125.98mils)	NPTH	Round	L1 (Sig, PWR) - L2 (Sig, PWR)	Mechanical
	Total 4					

BATTERY_INTERFACE_BOARD Fabrication Document

Drill Drawing L1 - L2 (Scale 1:1)

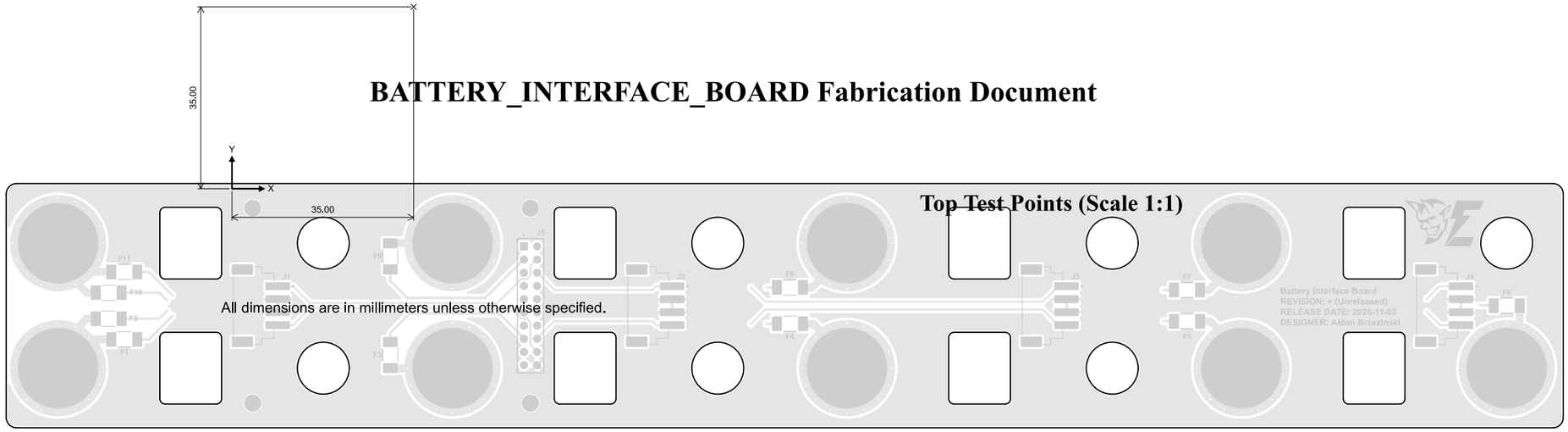


Comments:	Company: Blue Devil Motor Sports			Variant: CHECKED	Git Hash: e247924
	Board Name: BATTERY_INTERFACE_BOARD			Project Name: Battery Interface Board	
Sheet Title: Drill Drawing (L1 - L2)	File Name: battery_interface_board.kicad_pcb	Designer: Aidan Brzezinski		Date: 2026-10-12	Revision: + (Unreleased)
Sheet Path:		Reviewer:		Size: A3	Sheet: 4 of 8

Ref.	Net	X [mm]	Y [mm]

Ref.	Net	X [mm]	Y [mm]

BATTERY_INTERFACE_BOARD Fabrication Document

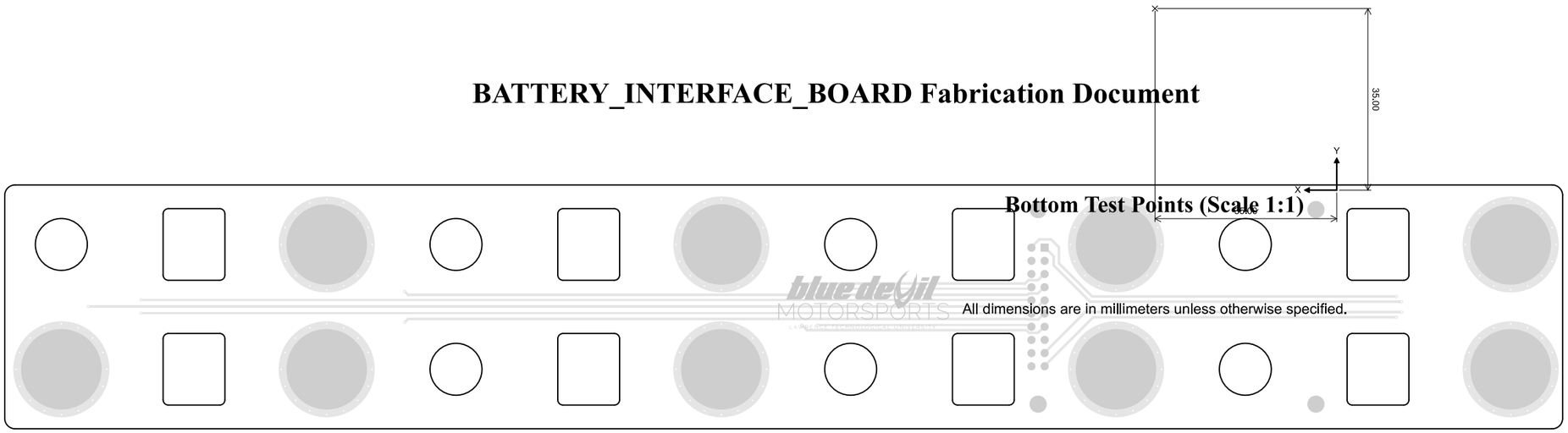


Top Test Points (Scale 1:1)

Comments:	Company: **Blue Devil Motor Sports**		Variant: CHECKED	Git Hash: e247924
	Board Name: **BATTERY_INTERFACE_BOARD**		Project Name: **Battery Interface Board**	
Sheet Title: Top Test Points (Scale 1:1)	File Name: battery_interface_board.kicad_pcb	Designer: Aidan Brzezinski	Date: 2026-10-12	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **5** of **8**

Ref.	Net	X [mm]	Y [mm]

BATTERY_INTERFACE_BOARD Fabrication Document



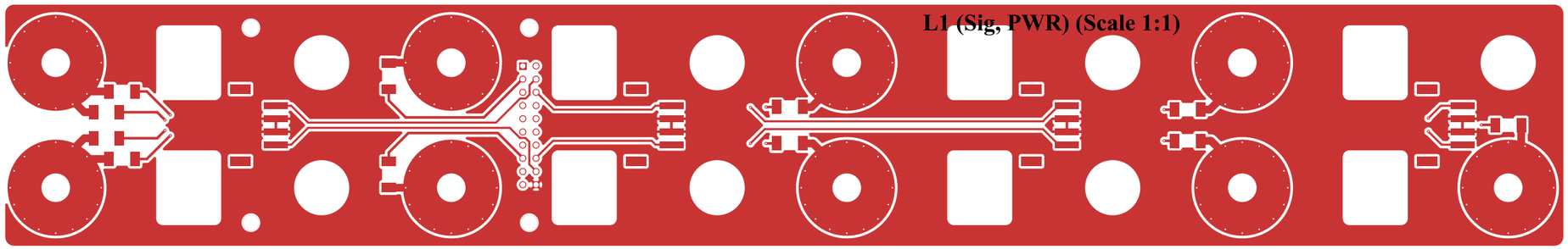
Bottom Test Points (Scale 1:1)

All dimensions are in millimeters unless otherwise specified.

Comments:	Company: **Blue Devil Motor Sports**		Variant: CHECKED	Git Hash: e247924
	Board Name: **BATTERY_INTERFACE_BOARD**		Project Name: **Battery Interface Board**	
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: battery_interface_board.kicad_pcb	Designer: Aidan Brzezinski	Date: 2026-10-12	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **6** of **8**

BATTERY_INTERFACE_BOARD Fabrication Document

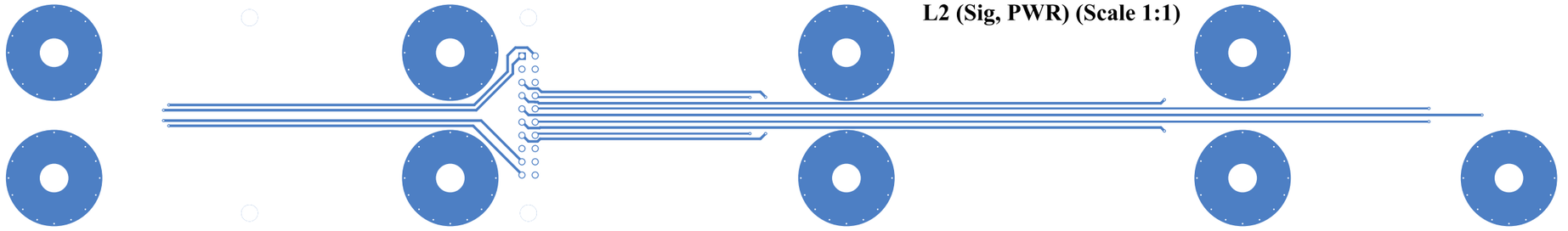
L1 (Sig, PWR) (Scale 1:1)



Comments:	Company: **Blue Devil Motor Sports**		Variant: CHECKED	Git Hash: c247924
	Board Name: **BATTERY_INTERFACE_BOARD**		Project Name: **Battery Interface Board**	
Sheet Title: L1 (Sig, PWR) (Scale 1:1)	File Name: battery_interface_board.kicad_pcb	Designer: Aidan Brzezinski	Date: 2026-10-12	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **7** of **8**

BATTERY_INTERFACE_BOARD Fabrication Document

L2 (Sig, PWR) (Scale 1:1)



Comments:	Company: **Blue Devil Motor Sports**		Variant: CHECKED	Git Hash: e247924
	Board Name: **BATTERY_INTERFACE_BOARD**		Project Name: **Battery Interface Board**	
Sheet Title: L2 (Sig, PWR) (Scale 1:1)	File Name: battery_interface_board.kicad_pcb	Designer: Aidan Brzezinski	Date: 2026-10-12	Revision: + (Unreleased)
Sheet Path:		Reviewer:	Size: **A3**	Sheet: **8** of **8**